

DBDOME Standard Operating Procedures

Daily operations, alert handling, change control and escalation

Product	DBDOME Database Security and Monitoring Platform
Applies to	DBDOME 2.02.x
Document version	1.0
Date	2026-08-23
Audience	Security operations, DBAs, platform administrators
Review cycle	Annual, or on any major version change
Classification	Customer deliverable

DBDOME Standard Operating Procedures

1. Purpose and scope

This document defines the routine procedures for operating DBDOME: daily checks, alert handling, onboarding of monitored databases, change control, backup and escalation.

It assumes DBDOME is already installed and verified per the *DBDOME Installation Guide*. It does not cover installation or upgrade, which that document owns.

A note on what this platform is. DBDOME is a detective control. It observes, records and alerts. It is one control among many and does not replace backups, patching, access governance or trained staff. Procedures below are written on that assumption.

2. Roles and responsibilities

Role	Responsibility
Platform Administrator	Availability of DBDOME itself; upgrades; backups; capacity
Security Analyst	Triage and disposition of alerts; tuning; incident escalation
DBA	Target-database credentials and access; validating findings against the database
Data Protection Officer / Legal	Lawful basis for monitoring; retention policy; access to monitoring data
Approver	Authorizes enabling of any enforcement feature (blocking, masking, suppression)

Enforcement features are **never** enabled by the person who requests them. Separation between requester and approver is required — see Section 8.

3. Daily operations

3.1 Morning platform health check

Perform each working day, ideally before the analyst shift starts.

Step 1 — Services

```
Get-Service DBDOME_dbanalytics, DBDOME_web, DBDOME_Grafana, DBDOME_Ollama, postgresql-x64-18 |  
Format-Table Name, Status, StartType
```

All must be **Running**. A stopped `DBDOME_dbanalytics` means **no collection and no alerting is occurring at all** — treat as a P1 incident, not a routine finding.

Step 2 — Collection is live

```
SELECT max(entry_date) AS last_collection  
FROM monitoring.general_metric_metadata_results;
```

Must be recent and advancing between checks. A running service with a static timestamp is a failure that service state does not reveal.

Step 3 — Per-server freshness

```
SELECT server, max(entry_date) AS last_seen  
FROM monitoring.general_metric_metadata_results  
GROUP BY server ORDER BY last_seen;
```

Investigate any server whose `last_seen` is materially behind the others.

Known failure mode — a dead target starves the whole sweep. An unreachable target consumes the collection sweep waiting on timeouts. Metric families ordered later in the sweep silently stop updating while earlier ones stay current. The symptom is *partial* staleness, which looks like a metric problem but is actually one unreachable server. Always check this per-server view before investigating an individual metric.

Step 4 — Open alerts

Review the Open Alerts dashboard. Confirm the overnight queue has been triaged and nothing is ageing beyond the SLA in Section 5.4.

Step 5 — Internal health

Review internal health alerts (disk, CPU, memory, service-down). DBDOME monitors itself; these alerts are about the DBDOME host, not the targets.

3.2 Record

Log the check in the operations record: date, operator, outcome, any action taken. This record is often the evidence auditors ask for.

4. Weekly and monthly operations

4.1 Weekly

Task	Notes
Review alert volume and top rules	A rule producing constant alerts is a tuning problem, not a security finding
Review false-positive rate per rule	Feed into Section 6 tuning
Confirm backups completed and are restorable	A backup never test-restored is not a backup
Check disk headroom on the DBDOME host and PostgreSQL volume	
Review agent verdict quality	Section 7
Confirm all expected servers are reporting	Section 3.1 step 3

4.2 Monthly

Task	Notes
Review user access to DBDOME itself	Monitoring data is sensitive; restrict accordingly
Review retention against policy	
Review and re-approve any enabled enforcement feature	Section 8
Patch review — DBDOME, PostgreSQL, Grafana, OS	
Capacity trend review	Partition growth, disk, RAM
Test the escalation path	Confirm on-call contacts still valid

4.3 Quarterly

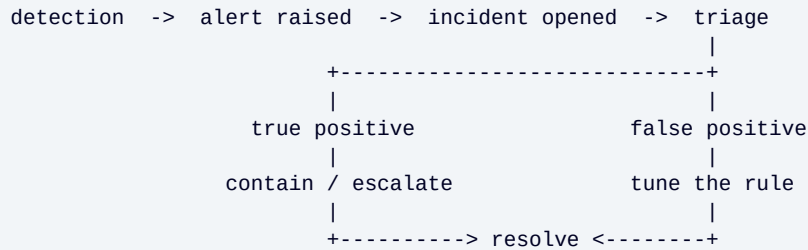
- Restore a database backup to a scratch host and verify integrity end to end.
- Review the monitoring scope against the lawful-basis assessment — scope drift

is common and is a compliance exposure.

- Review this SOP for accuracy.

5. Alert handling

5.1 Lifecycle



Alerts are written to `alerts.alert_log`. Open incidents live in `alerts.alert_incidents`. Incidents are resolved manually by the customer, or automatically where auto-resolve is configured for that rule.

5.2 Triage procedure

For each new alert:

1. **Read the alert record.** Note the rule, server, login, client program, and the statement that triggered it.
2. **Read the agent verdict if present** — the `reason` field on the alert metadata. Treat it as a starting hypothesis, never as a conclusion. See Section 7.
3. **Establish whether the activity is expected.** Check the statement against known application behaviour for that server, and check whether the login and client program are ones that normally appear.
4. **Confirm against the database itself.** Independently verify with the DBA rather than relying solely on DBDOME's record.
5. **Decide:** true positive, false positive, or benign-but-noteworthy.
6. **Record the disposition and the reason**, then resolve the incident with the appropriate resolution type.

5.3 Evidence handling

Where an alert may become the basis of a disciplinary, legal or regulatory process:

- Preserve the alert record and supporting metric rows **before** any retention job can age them out.
- Export and store them under the organization's evidence procedure.
- Do not rely on the AI-generated reason as evidence. It is a generated explanation, not a record of fact, and must not be presented as one.
- Involve Legal and the DPO early.

5.4 Response targets

Set locally; the following is a reasonable default.

Severity	Acknowledge	Triage complete
Critical	15 minutes	1 hour
High	1 hour	4 hours
Medium	4 hours	1 working day
Low	1 working day	5 working days

Alert volume that makes these targets unachievable is a tuning problem. Escalate it as such rather than allowing the queue to age — an unread queue is indistinguishable from no monitoring.

6. Rule tuning

6.1 When to tune

Tune when a rule produces repeated alerts that triage consistently dismisses. Persistent false positives are corrosive: they train analysts to dismiss the rule without reading it, which is worse than not having the rule.

6.2 Procedure

1. Quantify: how many alerts, over what period, what proportion dismissed.
2. Identify the discriminator that separates real findings from noise.
3. Propose the change (threshold, exclusion, schedule) with the evidence.
4. **Obtain approval** — tuning reduces detection coverage and is a change to a security control.
5. Apply in a change window.
6. Record: what changed, why, who approved, when.
7. Re-review after two weeks.

6.3 Constraints

Do not silence a rule by disabling it where narrowing it would do. A disabled rule looks identical to a rule that never fires, and the reason is lost within weeks.

Exclusions are the most dangerous tuning tool. An exclusion for a login or program is exactly what an attacker would want. Every exclusion must be reviewed at the monthly review and must have a stated owner and expiry.

Adding detection or resolution steps to a rule multiplies collection cost — each additional detection step multiplies the metric rows written, and each resolution step causes repeated execution against the customer database. Review the cost of a rule change, not just its logic.

7. AI security-agent verdicts

7.1 What the verdict is

The agent compares the triggering query against activity previously observed on that server and writes a plain-language reason onto the alert.

7.2 How to use it

Use it to: orient quickly, spot precedent you would otherwise have to search for, and prioritize a queue.

Do not use it to: close an alert without independent checking, justify a disciplinary or legal action, or serve as evidence.

Verdicts are **not currently trustworthy enough to suppress alerts**. Benign queries have been classified as alerts, and reasons have contained factual slips about the query being described. Annotate-only is the supported mode.

7.3 Verdict quality review — weekly

```
SELECT decided_by, count(*), round(avg(elapsed_ms))
FROM alerts.security_agent_verdict
GROUP BY decided_by;
```

`decided_by='model'` is a genuine verdict. Any other value means the agent failed open — alerts still fired, without a reason attached.

Sample a handful of `model` verdicts each week and compare them against the analyst's own disposition. Record agreement. **If agreement is poor, say so and stop relying on the verdicts** — a confidently wrong reason is worse than none, because it anchors the analyst.

7.4 If verdicts stop appearing

Work through, in order:

1. `SECURITY_AGENT_ENABLED=true` in `.env`?
2. Model reachable? Run the agent check per the Installation Guide, Section 8.3.
3. Sweep registered and running? Check `metrics.registered_processes`, and check

the service log for `Unknown process name`.

4. Ollama port owned by SYSTEM? Installation Guide, Section 8.4.
5. Is collection healthy? With no precedent data the agent has nothing to compare

against.

8. Enforcement features — blocking, masking, suppression

DBDOME can terminate sessions, block logins, mask data and suppress alerts. **These are disruptive and are disabled by default.**

8.1 Mandatory conditions before enabling

- Documented business case
- Tested in a non-production environment against realistic traffic
- Dry-run mode observed for a defined period with the actions it *would*

have taken reviewed

- Blast radius understood — what breaks if it fires wrongly
- Approved by someone other than the requester
- Rollback procedure written and tested
- Application owners notified

8.2 Ongoing

- Review every enforcement action taken, weekly.
- Re-approve the feature monthly.
- Any wrongful block is an incident: record it, review the rule, and consider

returning the feature to dry-run.

Alert suppression by AI verdict is the highest-risk setting in the platform. It allows a model to decide that a security alert is not worth telling anyone about. Given the accuracy limits in Section 7.2, it should remain off.

9. Onboarding a monitored database

1. **Confirm authorization in writing** to monitor that system.
2. **Confirm lawful basis** — notification, consultation or DPIA as required.
3. Create a **read-only** monitoring login on the target. Do not grant

`sysadmin`, `DBA` or ownership.

4. Confirm network reachability from the DBDOME host.

5. Register the server in DBDOME with vendor, host, port and **the specific database**.

6. Verify collection within 30 minutes:

```
SELECT max(entry_date) FROM monitoring.general_metric_metadata_results
WHERE server = 'new server';
```

7. Observe for a baseline period before enabling alerting, so normal traffic is represented.

8. Record the server in the monitored-systems inventory with its owner.

Common onboarding failure. Login failed for <user> (18456) together with 4060 Cannot open database None is a **NULL database in the configuration**, not a credential problem. Set the database explicitly. Hours are routinely lost re-issuing credentials that were never wrong.

9.1 Offboarding

Remove the server from DBDOME, revoke the monitoring login on the target, and record the removal. Confirm no alert rule still references it.

10. Backup and recovery

10.1 What must be backed up

Item	Why
DBDOME PostgreSQL database	All configuration, metrics, alerts, incidents
bin\env	Connection settings and the encryption key
bin\internal\security_agent\rules.json	Local rule customization; replaced by upgrades
Grafana grafana.db	Dashboards and their customizations
Any custom report templates	

DBDOME_SECRET_KEY **deserves separate treatment**. Stored credentials are encrypted with it. **Back it up somewhere other than the DBDOME host**. Losing it means every stored target credential must be re-entered by hand, and a host failure that takes the key with it turns a restore into a re-onboarding exercise.

10.2 Schedule

Item	Frequency	Retention
Full database backup	Daily	Per policy
Configuration files	On change, and weekly	3 versions minimum
Encryption key	On change	Permanent, offsite, access-controlled
Restore test	Quarterly	Record the result

10.3 Recovery

1. Restore the database from backup.
2. Restore bin and env.
3. Confirm ORG_IP matches the recovered host's LAN IP — if the host changed, this must be corrected or dashboards and ingest will fail.
4. Start services and run the full verification in the Installation Guide,

Section 7.

5. Confirm collection resumed and stored credentials still decrypt.

11. Change control

Any of the following is a change to a security control and requires the process below:

- Enabling, disabling or modifying a detection rule
- Adding or removing an exclusion
- Changing a threshold
- Enabling any enforcement feature
- Upgrading DBDOME
- Adding or removing a monitored server
- Changing retention

Process: request → impact assessment → approval → scheduled window → implementation → verification → record.

Record, at minimum: what changed, why, who approved it, when it was applied, how it was verified, and how to reverse it.

12. Escalation

Situation	Escalate to	Timeframe
Confirmed data exfiltration or ransomware indicators	Security incident response	Immediate
DBDOME collection stopped	Platform Administrator	Immediate
Suspected insider misuse	Security lead + HR + Legal	Immediate, before confronting anyone
Wrongful block or mask affecting production	Platform Administrator + application owner	Immediate
Alert queue beyond SLA	Security lead	Same day
Repeated false positives on one rule	Security lead for tuning approval	Weekly review
Suspected DBDOME defect	support@dbexpert.ai	Per support agreement

Maintain a current on-call contact list. Test it monthly — an escalation path that has never been exercised usually fails on first use.

13. Health indicators

Track these. They describe whether monitoring is actually working, which service state alone does not.

Indicator	Target
Collection freshness (all servers)	Within one collection interval
Servers reporting vs. servers registered	100%
Alerts triaged within SLA	> 95%
False-positive rate	Trending down
Open incidents ageing beyond SLA	0
Backup success	100%
Restore test	Passed within the quarter
Agent verdicts decided_by= 'model'	Stable; investigate drops

Indicator	Target
Analyst/agent agreement on sampled verdicts	Recorded weekly

14. Records to retain

Record	Retain
Daily health check log	12 months
Alert dispositions	Per policy; typically 12–24 months
Change records	Life of the system + 12 months
Approvals for enforcement features	Life of the system
Exclusion register with owner and expiry	Life of the system
Backup and restore-test results	12 months
Monitored-systems inventory with authorization	Current + history
Lawful-basis assessment	Per DPO policy

15. Appendix — quick command reference

Service state

```
Get-Service DBDOME_dbanalytics, DBDOME_web, DBDOME_Grafana, DBDOME_Ollama, postgresql-x64-18
```

Restart collection (requires elevation)

```
Restart-Service DBDOME_dbanalytics
```

Collection freshness

```
SELECT server, max(entry_date) FROM monitoring.general_metric_metadata_results
GROUP BY server ORDER BY 2;
```

Open incidents

```
SELECT * FROM alerts.alert_incidents WHERE resolved_at IS NULL ORDER BY 1 DESC;
```

Agent verdict summary

```
SELECT decided_by, count(*) FROM alerts.security_agent_verdict GROUP BY 1;
```

Scheduled processes

```
SELECT process_name, is_active FROM metrics.registered_processes ORDER BY 1;
```

Ollama port ownership

```
Get-NetTCPConnection -LocalPort 11434 -State Listen |
  ForEach-Object { (Get-Process -Id $_.OwningProcess).ProcessName }
```

Logs: C:\ProgramData\DBDOME\bin\logs\, and the log schema in the database.

16. Support

Technical support	support@dbexpert.ai
Licensing	legal@dbexpert.ai

Copyright (c) 2026 DBEXPERT. All rights reserved. DBDOME is a trademark of DBEXPERT.